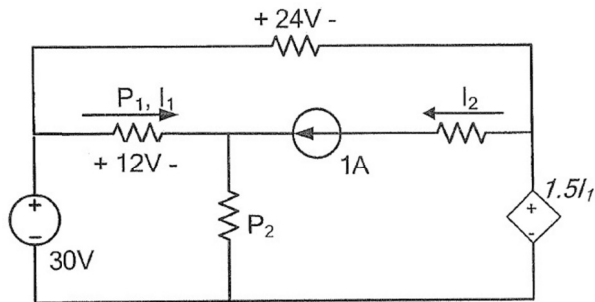


EBN 111: Tutorial Test 1 Memo (2012-02-24)

Kringbaan 1 / Circuit 1:



V1/Q1: Vir kringbaan 1, bepaal die drywing P_1 . / For circuit 1, determine the power P_1 .

$$-30 \text{ V} + 24 \text{ V} + 1.5 I_1 = 0 \text{ (KVL)}$$

Thus: $I_1 = 4 \text{ A}$.

$$P_1 = +12 \text{ V} * 4 \text{ A} = 48 \text{ W}$$

V2/Q2: Vir kringbaan 1, bepaal die drywing P_2 . / For circuit 1, determine the power P_2 .

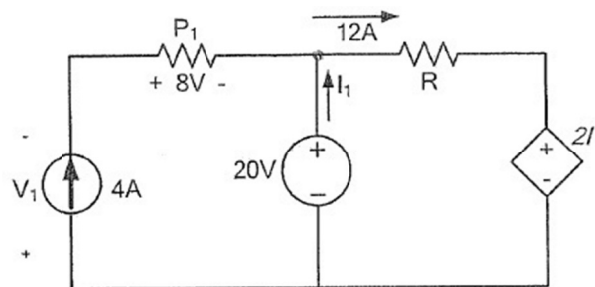
$$I_1 = 4 \text{ A}$$

$$\text{KCL: } I_1 + 1 \text{ A} = 5 \text{ A}$$

$$\text{KVL: } -30 \text{ V} + 12 \text{ V} = V_{P2} = 18 \text{ V}$$

$$P_2 = +18 \text{ V} * 5 \text{ A} = 90 \text{ W}$$

Kringbaan 2 / Circuit 2:



V3/Q3: Vir kringbaan 2, bereken V_1 . / For circuit 2, calculate V_1 .

$$V_1 + 8 + 20 = 0 \text{ (KVL)}$$

$$V_1 = -28 \text{ V}$$

V4/Q4: Vir kringbaan 2, bereken I_1 . / For circuit 2, calculate I_1 .

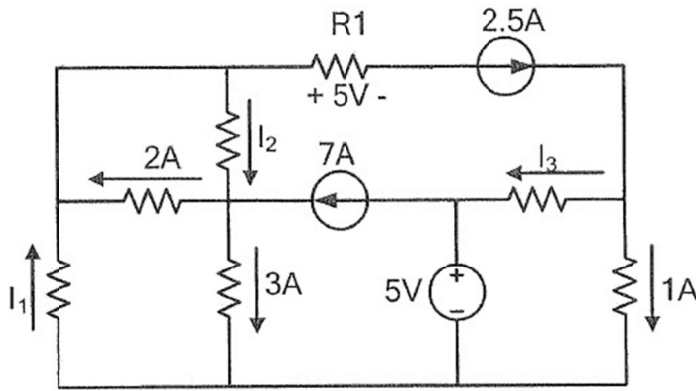
$$I_1 + 4 = 12 \text{ (Nodal analysis)}$$

$$I_1 = 8 \text{ A}$$

V5/Q5: Vir kringbaan 2, bereken die drywing P_1 . / For circuit 2, calculate the power P_1

$$P_1 = V \times I = 8 \times 4 = 32 \text{ W}$$

Kringbaan 3 / Circuit 3:



V6/Q6: Vir kringbaan 3, bereken I_2 . / For circuit 3, calculate I_2 .

$I_2 + 7 = 2 + 3$ (KCL)

$I_2 = -2 \text{ A}$

V7/Q7: Vir kringbaan 3, bereken die drywing geassosieer met die 5V bron. / For circuit 3, calculate the power associated with 5V source.

Denote the current passing through the 5V as I_5 . Therefore the power of the 5V source is,

$P_{5V} = 5 \times I_5 = 5I_5$

KCL: $7 + I_5 = I_3$; $I_5 = I_3 - 7$

Also: $2.5 = I_3 + 1$; $I_3 = 1.5 \text{ A}$

Therefore:

$I_5 = I_3 - 7$

$I_5 = 1.5 - 7$

$I_5 = -5.5 \text{ A}$

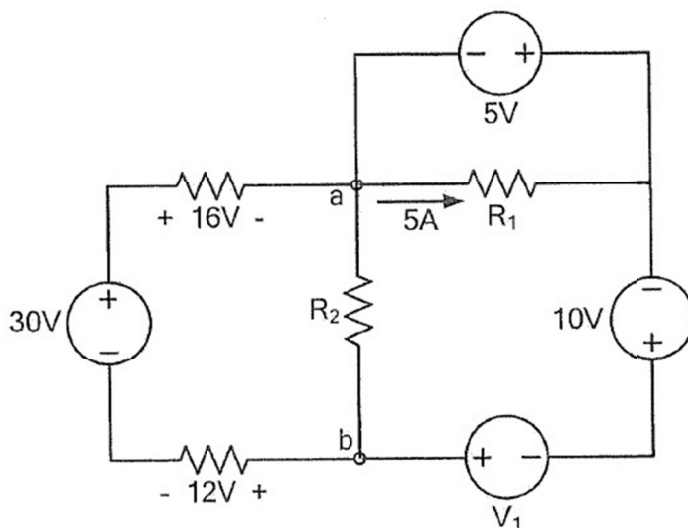
Finally:

$P_{5V} = 5 \times I_5$

$P_{5V} = 5 \times -5.5$

$P_{5V} = -27.5 \text{ W}$

Kringbaan 4 / Circuit 4:



V8/Q8: Vir kringbaan 4, bereken V_1 . / For circuit 4, determine V_1 .

KVL around outside: $-30 \text{ V} + 16 \text{ V} - 5 \text{ V} - 10 \text{ V} - V_1 + 12 \text{ V} = 0$; Thus $V_1 = -17 \text{ V}$

V9/Q9: Vir kringbaan 4, bepaal V_{ab} . / For circuit 4, determine V_{ab} .

KVL: $-30 \text{ V} + 16 \text{ V} + V_{ab} + 12 \text{ V} = 0$; Thus $V_{ab} = 2 \text{ V}$